

PROBEAT

PRObiotic Bacterial gEnome Analysis Toolkit

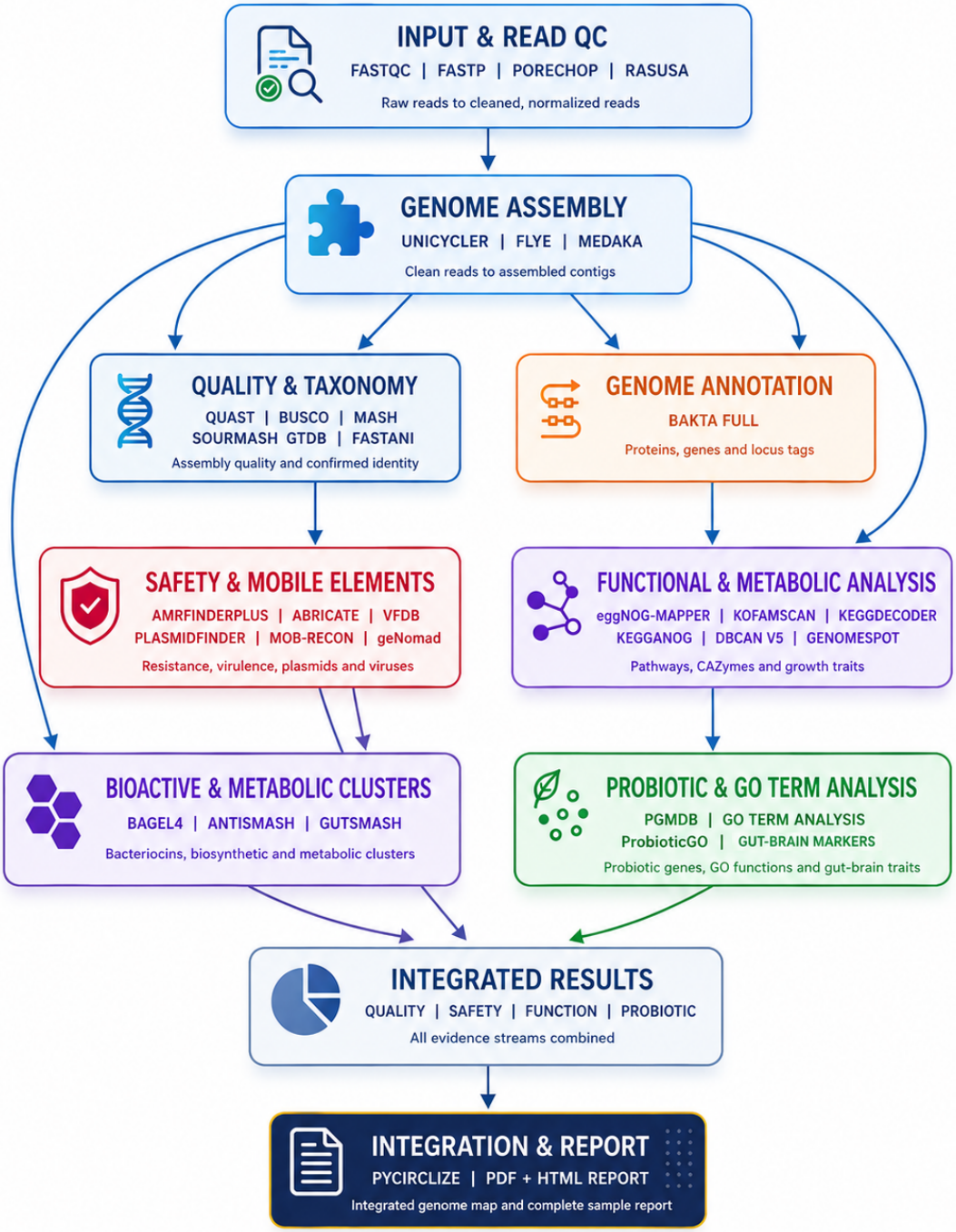
PRObiotic Bacterial gEnome Analysis Toolkit (PROBEAT)

Generated: 2026-06-28T19:41:54

PAISII HILENDARSKI UNIVERSITY OF PLOVDIV

115,161 Raw reads	**7** Contigs (>= 500 bp)	**2,994,811 bp** Genome size
45.43% GC	**Levilactobacillus brevis** Species	**2,776,991** N50

Workflow



Contents

- Overview and Workflow 2
- Per-sample Summary 4
- Flye Assembly Report: SRR30409709 5
- Bakta Report: SRR30409709 6
- Integrated Bacterial Genome Map: SRR30409709 7
- FastANI Species Confirmation: SRR30409709 8
- geNomad Plasmid and Virus Summary: SRR30409709 9
- MOB-recon Contig Classification: SRR30409709 10
- AMRFinderPlus & VFDB: SRR30409709 11
- ABRicate CARD & MEGARES: SRR30409709 12
- BAGEL4 Bacteriocin and RiPP Clusters: SRR30409709 13
- antiSMASH Biosynthetic Gene Clusters: SRR30409709 14
- dbCAN CAZyme Summary: SRR30409709 15
- dbCAN Prebiotic Carbohydrate Utilization: SRR30409709 16
- dbCAN CGC Substrate Predictions: SRR30409709 18
- dbCAN CGC Synteny Maps: SRR30409709 19
- PGMDB Probiotic Gene Summary: SRR30409709 21
- PGMDB Probiotic Gene Coverage: SRR30409709 22
- PGMDB Probiotic Genes: SRR30409709 23
- Gut-Brain Axis Markers: SRR30409709 30
- KEGGDecoder Functional Completeness 32
- KEGGaNOG Pathway Completeness (from eggNOG) 33
- Nutrient Uptake Potential (KEGG TC) - SRR30409709 34
- EC Enzyme Classification - SRR30409709 35
- COG Functional Categories 36
- GenomeSPOT Growth Condition Predictions: SRR30409709 37
- gutSMASH Metabolic Clusters: SRR30409709 38
- GO Enrichment: SRR30409709 39
- Genome GO Functional Landscape: SRR30409709 41
- Probiotic GO Categories: SRR30409709 42
- Results Directory Structure 43

Per-sample Summary

Sample SRR30409709 (ONT)

QC & Assembly

Raw reads/bases: 115,161 / 454,428,661
After adapter/QC reads/bases: 115,639 / 440,778,636; Q30 31.07% **(!)**
Adapter-trimmed reads/bases: 0 / 13,650,025
Rasusa: subsampled_by_coverage; NA -> NA reads
Coverage: 147.2x
Assembly: 7 contigs; 2,994,811 bp **(!)** ; GC 45.43% **(!)** ; N50 2,776,991
BUSCO: Complete 94.4%; single-copy 94.4%
BUSCO duplicated: 0.0%; fragmented 0.8%; missing 4.8%

Identification & Annotation

Taxonomy: sourmash Levilactobacillus brevis 74.5%; Mash Lactobacillus brevis
FastANI: 99.32% to Levilactobacillus brevis strain NPS-QW-145; PASS
Bakta: CDS 3,072; tRNA 100; rRNA 15; tmRNA 1; CRISPR 3
MOB-recon: 1 plasmid contigs; 1 reconstructed plasmids; 0 mobilizable
geNomad: 3 plasmid contigs; 7 virus/provirus contigs

Safety

AMRFinderPlus AMR hits: 0
PlasmidFinder replicon hits: 2
VFDB virulence-factor hits: 0

Functional & Probiotic Potential

KofamScan KO annotations: 1,389
KEGGDecoder functions present: 15 / 16
dbCAN CAZymes: 78
PGMDB: 176 unique; Vitamin biosynthesis 25, Acid stress 40, Bile resistance 28, Exopolysaccharide production 9
gutSMASH: 3 metabolic clusters; top match Arginine to hydrogen carbonate P. aeruginosa
GenomeSPOT: oxygen tolerant; optimum 20.1 C; pH 5.4
antiSMASH: 2 biosynthetic clusters; T3PKS, terpene-precursor
Gut-Brain: 51 primary markers; top traits GABA; acid tolerance, Gastric acid survival
GO Probiotic: 5 significant, 0 supportive probiotic GO categories; Antioxidant Capacity, Metal and Environmental Stress, Vitamin Production
BAGEL4 bacteriocin/RiPP clusters: 1

Flye Assembly Report: SRR30409709

ONT long-read contig statistics

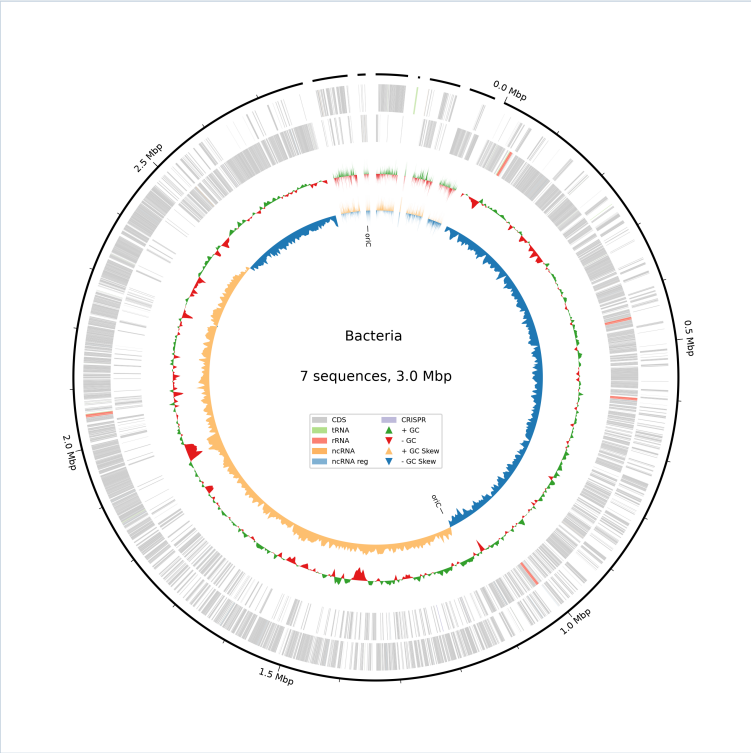
Contig	Length	Coverage	Circular	Repeat	Multiplicity	Alt group
contig_2	2,777,222	61	No	No	1	*
contig_3	57,216	238	Yes	No	2	*
contig_6	52,341	740	No	Yes	7	*
contig_7	50,157	180	Yes	No	1	*
contig_5	43,163	2285	Yes	Yes	21	*
contig_4	12,388	184	Yes	No	1	*
contig_8	2,552	99	Yes	Yes	1	*

Bakta Report: SRR30409709

Sequence(s):
Length: 2994811
Count: 7
GC: 45.4
N50: 2776991
N90: 2776991
N ratio: 0.0
coding density: 87.3

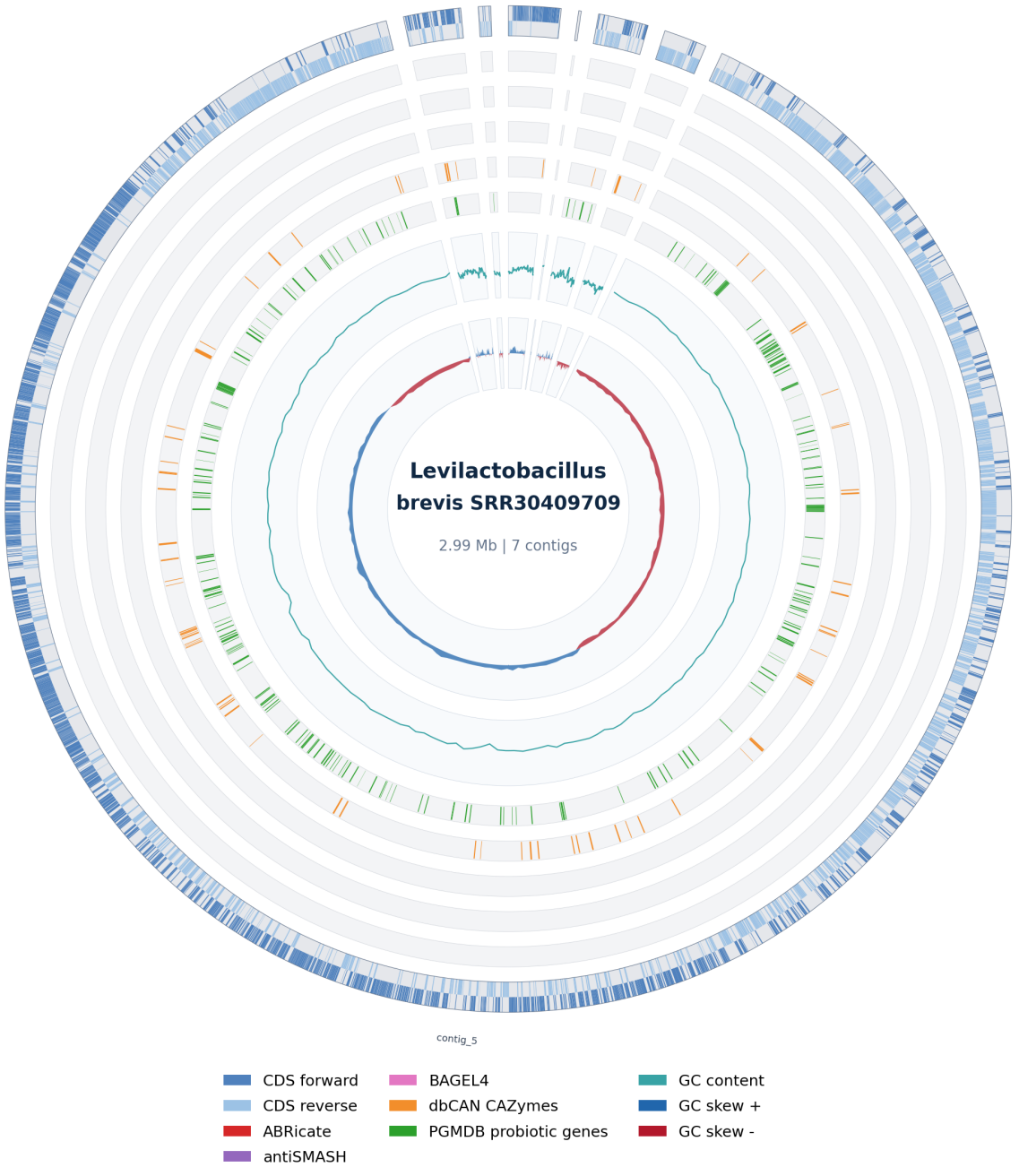
Annotation:
tRNAs: 100
tmRNAs: 1
rRNAs: 15
ncRNAs: 7
ncRNA regions: 27
CRISPR arrays: 3
CDSs: 3072
pseudogenes: 23
hypotheticals: 366
sORFs: 0
gaps: 0
oriCs: 2
oriVs: 0
oriTs: 0

Bakta:
Software: v1.12.0
Database: v6.0, full
DOI: 10.1099/mgen.0.000685
URL: github.com/oschwengers/bakta



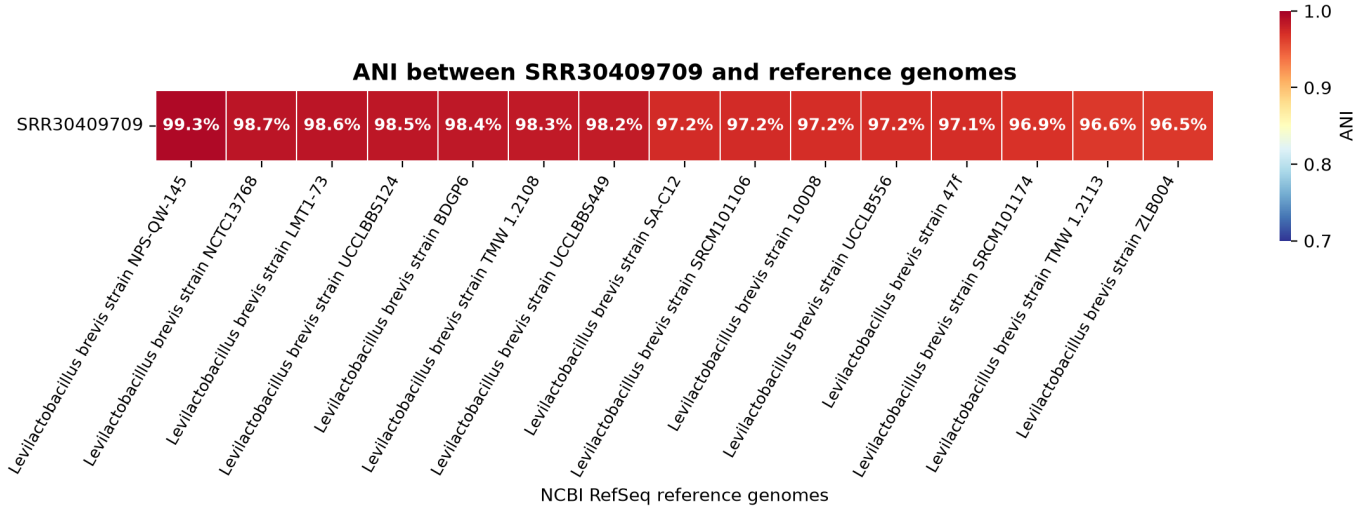
Integrated Bacterial Genome Map: SRR30409709

Bakta CDS, ABRicate, antiSMASH, BAGEL4, dbCAN CAZymes, PGMDB probiotic genes, GC content and GC skew



FastANI Species Confirmation: SRR30409709

Assumed species: Levilactobacillus brevis (from sourmash GTDB)



PASS: Best FastANI 99.32% > 95%.

The assembled strain is correctly assigned to Levilactobacillus brevis.

geNomad Plasmid and Virus Summary: SRR30409709

Plasmid sequences identified by geNomad; 3 contig(s) classified as plasmids

Contig	Length	Topology	Plasmid score	FDR	Hallmarks	Marker enrichment
contig_4	12388	DTR	0.9938	NA	0	2.0515
contig_3	57217	No terminal repeats	0.9922	NA	2	9.5266
contig_7	50158	No terminal repeats	0.9911	NA	1	4.9178

geNomad Virus Summary

Virus sequences identified by geNomad; 7 contig(s) classified as viruses

Contig	Length	Topology	Virus score	FDR	Hallmarks	Marker enrichment
contig_2|provirus_2737339_27769	39653	Provirus	0.9673	NA	12	46.2037
contig_6	52342	No terminal repeats	0.9648	NA	14	60.9379
contig_5	43163	No terminal repeats	0.9641	NA	13	52.5257
contig_2|provirus_2325485_23662	40781	Provirus	0.9629	NA	11	38.4183
contig_2|provirus_2_40960	40959	Provirus	0.9560	NA	8	49.2380
contig_2|provirus_183491_236444	52954	Provirus	0.9558	NA	11	48.7135
contig_2|provirus_2075391_21239	48536	Provirus	0.9525	NA	16	61.2936

MOB-recon Contig Classification: SRR30409709

Only contigs classified as plasmids are shown; 1 plasmid contig(s)

Contig	Length	Rep types	Relaxases	Mobility	Nearest neighbor
contig_4	12388	rep_cluster_1255	-	-	Lactobacillus plantarum

MOB-recon Reconstructed Plasmids

Reconstructed plasmid groups and predicted host range; 1 group(s)

Plasmid	Contigs	Length	Rep types	Mobility	Nearest neighbor	Host range
SRR30409709:AB30	1	12388	rep_cluster_1255	non-mobilizable	Lactobacillus plantarum	Lactobacillales

AMRFinderPlus & VFDB: SRR30409709

No antimicrobial resistance genes detected.

No virulence-factor database matches detected.

ABRicate CARD & MEGARES: SRR30409709

No CARD antimicrobial resistance genes detected.

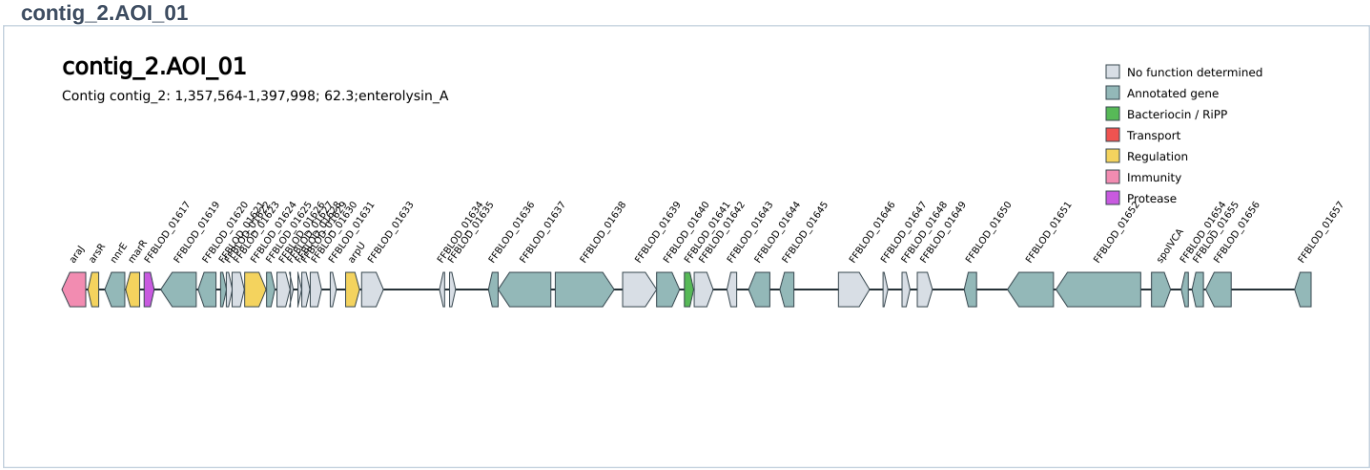
No MEGARES antimicrobial resistance genes detected.

BAGEL4 Bacteriocin and RiPP Clusters: SRR30409709

Areas of Interest detected by BAGEL4

AOI	Start	End	Class	Sequence
contig_2.AOI_01	1357564	1397998	Class III	62.3;enterolysin_A

contig_2.AOI_01



antiSMASH Biosynthetic Gene Clusters: SRR30409709

Predicted biosynthetic gene clusters with product type and gene count

Region	Location	Length	Products	Genes
1	2,261,376-2,302,545	41,170 bp	T3PKS	49
2	2,639,395-2,660,297	20,903 bp	terpene-precursor	25

dbCAN CAZyme Summary: SRR30409709

CAZymes supported by dbCAN consensus Recommend Results; 78 recommended genes

Class	Description	Consensus genes
GH	Glycoside Hydrolases	46
GT	Glycosyltransferases	27
PL	Polysaccharide Lyases	0
CE	Carbohydrate Esterases	3
AA	Auxiliary Activities	0
CBM	Carbohydrate-Binding Modules	5

dbCAN Prebiotic Carbohydrate Utilization: SRR30409709

Detected CAZy families matched against data/databases/probiotic_cazy.txt

Prebiotic substrate / group	Families	Bakta locus tags	Typical enzymes / functions
GOS / beta-galactooligosaccharides	GH2, GH42	FFBLOD_01117, FFBLOD_01230, FFBLOD_01441, FFBLOD_03136, FFBLOD_03148	beta-galactosidase; GOS/lactose hydrolysis
Lactose / lactose-derived oligosaccharides	GH1, GH2, GH42	FFBLOD_00843, FFBLOD_01117, FFBLOD_01230, FFBLOD_01441, FFBLOD_03133, FFBLOD_03136, FFBLOD_03148	beta-galactosidase; lactose hydrolysis; galactoside utilization
alpha-GOS / raffinose-family oligosaccharides (raffinose, stachyose , verbascose)	GH36	FFBLOD_02064, FFBLOD_03134	alpha-galactosidase; RFO hydrolysis
Sucrose-related oligosaccharides	GH13, GH36	FFBLOD_01047, FFBLOD_01555, FFBLOD_02064, FFBLOD_03134	Sucrose phosphorylase/hydrolase; alpha-glucosidase; fructosidase; alpha-galactosidase
XOS / xylooligosaccharides	GH120, GH3, GH43 , GH51, GH8	FFBLOD_00980, FFBLOD_01119, FFBLOD_01366, FFBLOD_01417, FFBLOD_01478, FFBLOD_01504, FFBLOD_01848, FFBLOD_01855, FFBLOD_02072, FFBLOD_02074	beta-xylosidase; exo-oligoxylanase; alpha-L-arabinofuranosidase
Arabinoxylan / AXOS	GH3, GH43, GH51	FFBLOD_00980, FFBLOD_01119, FFBLOD_01366, FFBLOD_01417, FFBLOD_01504, FFBLOD_01855, FFBLOD_02072, FFBLOD_02074	alpha-L-arabinofuranosidase; beta-xylosidase; xylanase; feruloyl/acetyl esterases
Arabinan / arabino-oligosaccharides (AOS)	GH1, GH127, GH3 , GH43, GH51	FFBLOD_00843, FFBLOD_00980, FFBLOD_01119, FFBLOD_01366, FFBLOD_01417, FFBLOD_01504, FFBLOD_01855, FFBLOD_02072, FFBLOD_02074, FFBLOD_02079, FFBLOD_03133	alpha-/beta-L-arabinofuranosidase; arabinan side-chain degradation
Xylan polymer	GH120, GH43, GH8	FFBLOD_00980, FFBLOD_01366, FFBLOD_01417, FFBLOD_01478, FFBLOD_01848, FFBLOD_02074	Endo-xylanase; beta-xylosidase; arabinofuranosidase; xylan debranching
Mannan / galactomannan / MOS	GH2, GH36	FFBLOD_01230, FFBLOD_01441, FFBLOD_02064, FFBLOD_03134, FFBLOD_03136	beta-mannanase; beta-mannosidase; alpha-galactosidase; galactomannan side-chain removal
Locust bean gum / guar gum / plant galactomannan	GH2, GH36	FFBLOD_01230, FFBLOD_01441, FFBLOD_02064, FFBLOD_03134, FFBLOD_03136	Endo-beta-mannanase; alpha-galactosidase; beta-galactosidase
Starch / resistant starch / maltodextrin	GH13, GH31, GH65	FFBLOD_00600, FFBLOD_00846, FFBLOD_00995, FFBLOD_01047, FFBLOD_01231, FFBLOD_01555, FFBLOD_01563, FFBLOD_02340	alpha-amylase; alpha-glucosidase; pullulanase; 4-alpha-glucanotransferase; starch-binding modules
Isomaltooligosaccharides / alpha-glucans	GH13, GH31	FFBLOD_00600, FFBLOD_00846, FFBLOD_01047, FFBLOD_01231, FFBLOD_01555	alpha-glucosidase; dextranase/isomaltodextranase; glucansucrase; 4, 6-alpha-glucanotransferase
Pectin side chains: galactan / arabinogalactan	GH2, GH43, GH51	FFBLOD_00980, FFBLOD_01230, FFBLOD_01366, FFBLOD_01417, FFBLOD_01441, FFBLOD_01504, FFBLOD_02072, FFBLOD_02074, FFBLOD_03136	Endo-beta-1,4-galactanase; beta-galactosidase; arabinofuranosidase
Type I/II arabinogalactan	GH127, GH43, GH51	FFBLOD_00980, FFBLOD_01366, FFBLOD_01417, FFBLOD_01504, FFBLOD_02072, FFBLOD_02074, FFBLOD_02079	beta-galactanase; alpha-/beta-arabinofuranosidase
beta-glucans: oat/barley mixed-linkage beta-glucan	GH3	FFBLOD_01119, FFBLOD_01855	Endo-beta-glucanase; beta-glucosidase; beta-glucan-binding modules

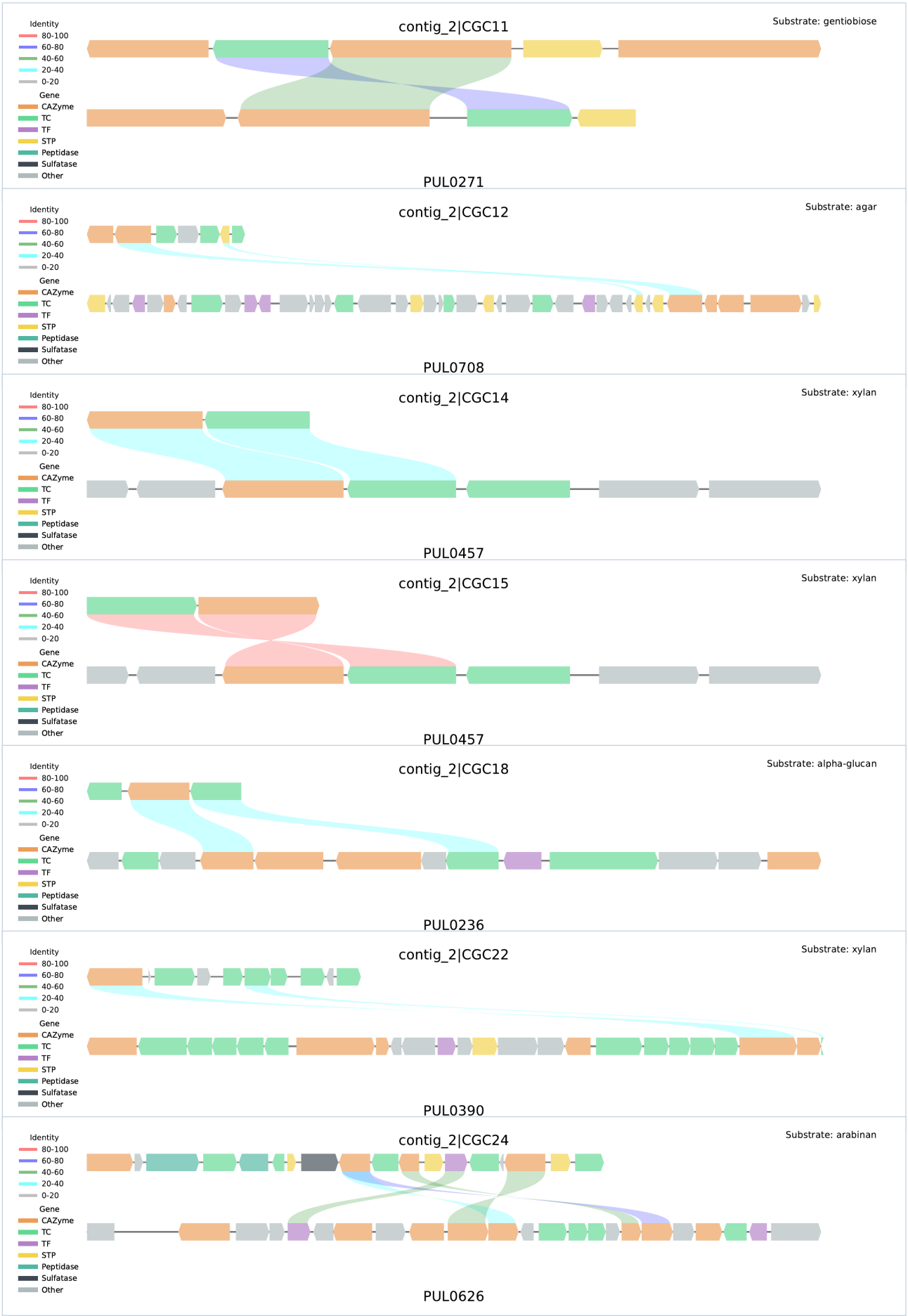
Prebiotic substrate / group	Families	Bakta locus tags	Typical enzymes / functions
Laminarin / fungal beta-1,3-glucan	GH3, GH30	FFBLOD_01115, FFBLOD_01119, FFBLOD_01498, FFBLOD_01855	beta-1,3-glucanase; beta-glucosidase
Cellobiose / cello-oligosaccharides	GH1, GH3	FFBLOD_00843, FFBLOD_01119, FFBLOD_01855, FFBLOD_03133	beta-glucosidase; cellobiose phosphorylase; cello-oligosaccharide utilization

dbCAN CGC Substrate Predictions: SRR30409709

CAZyme Gene Clusters with predicted substrates from dbCAN-PUL

CGC#	Location	Length	CAZymes	PULID	Substrate	Bitscore
CGC5	contig_2:437954-444948	6995 bp	2	PUL0591	glycosaminoglycan	393.0
CGC6	contig_2:533379-540240	6862 bp	2	PUL0648	xylan	333.0
CGC8	contig_2:685045-688741	3697 bp	1	PUL0611	alpha-glucan	938.0
CGC11	contig_2:815522-823570	8049 bp	3	PUL0271	gentiobiose	1068.0
CGC12	contig_2:935742-946595	10854 bp	2	PUL0708	agar	338.3
CGC14	contig_2:1085656-1088646	2991 bp	1	PUL0457	xylan	570.0
CGC15	contig_2:1140578-1143695	3118 bp	1	PUL0457	xylan	1770.0
CGC18	contig_2:1295035-1299214	4180 bp	1	PUL0236	alpha-glucan	286.9
CGC22	contig_2:1598148-1609279	11132 bp	1	PUL0390	xylan	330.0
CGC24	contig_2:1805389-1830173	24785 bp	4	PUL0626	arabinan	2186.0
CGC36	contig_3:13090-18401	5312 bp	3	PUL0561	alpha-galactan	3176.0
CGC37	contig_3:29016-33164	4149 bp	1	PUL0007	galactan	989.0

dbCAN CGC Synteny Maps: SRR30409709




Identity
80-100
60-80
40-60
20-40
0-20
Gene
CAZyme
TC
TF
STP
Peptidase
Sulfatase
Other
contig_2|CGC5
Substrate: glycosaminoglycan
PUL0591
contig_2|CGC6
Substrate: xylan
PUL0648
contig_2|CGC8
Substrate: alpha-glucan
PUL0611
contig_3|CGC36
Substrate: alpha-galactan
PUL0561
contig_3|CGC37
Substrate: galactan
PUL0007

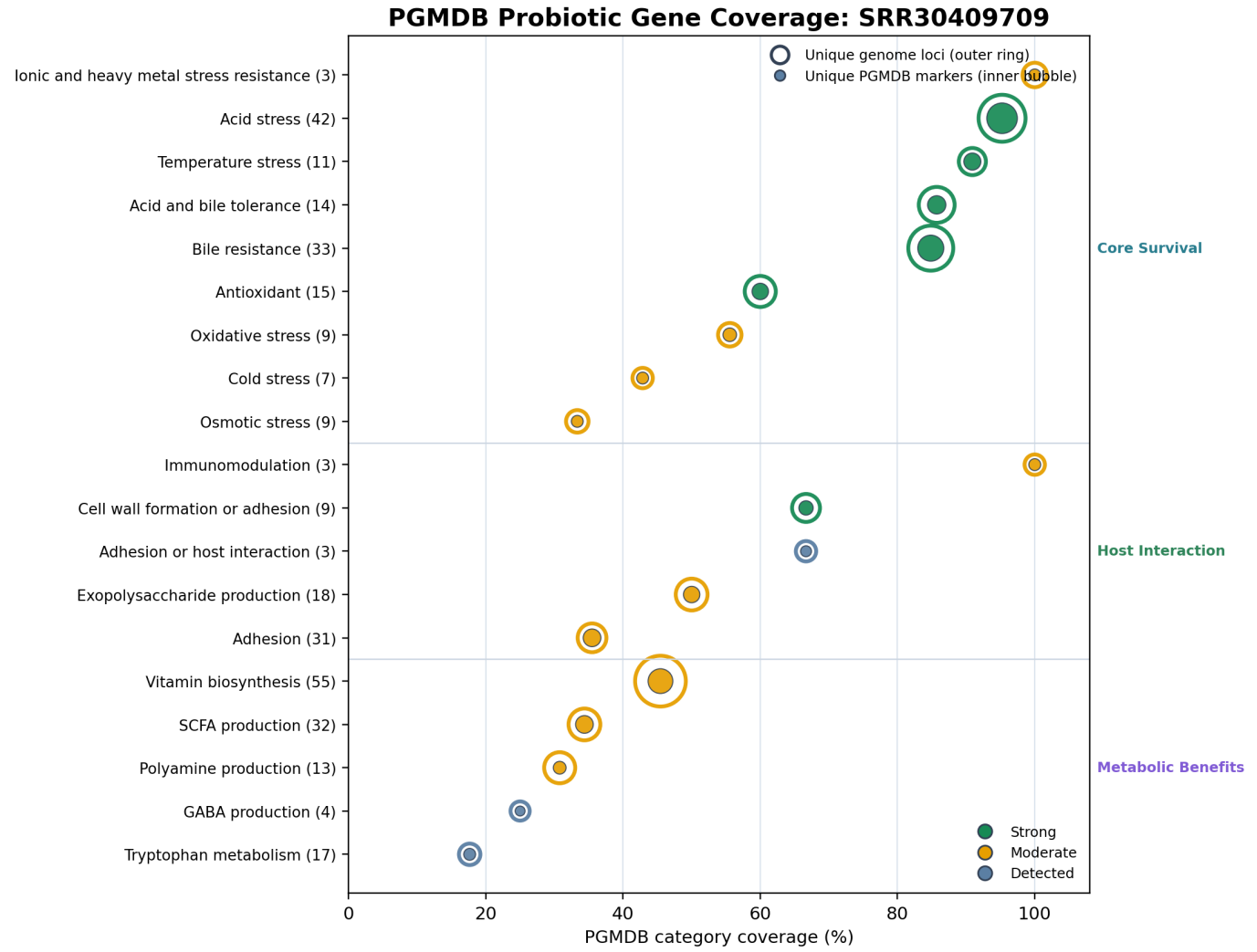
PGMDB Probiotic Gene Summary: SRR30409709

Curated probiotic-potential marker matches; presence does not demonstrate activity

Probiotic function	Hits	Unique markers
total_hits	276	176
Vitamin biosynthesis	50	25
Acid stress	42	40
Bile resistance	38	28
Acid and bile tolerance	22	12
Exopolysaccharide production	16	9
SCFA production	16	11
Antioxidant	15	9
Polyamine production	15	4
Adhesion	11	10
Cell wall formation or adhesion	11	6
Temperature stress	10	10
Ionic and heavy metal stress resistance	7	3
Oxidative stress	6	5
Osmotic stress	5	3
Tryptophan metabolism	4	3
Adhesion or host interaction	3	2
Cold stress	3	3
Immunomodulation	3	3
GABA production	2	1
Intestinal persistence	2	1
Proteases	2	2
Adhesions	1	1
Heat stress	1	1
Heat-shock stress	1	1
Nitrite stress	1	1
Stress adaptation	1	1
Stress response	1	1

PGMDB Probiotic Gene Coverage: SRR30409709

Only categories represented by at least 3 unique markers in PGMDB are shown



PGMDB Probiotic Genes: SRR30409709

PGMDB genes searched first in Bakta, then in eggNOG when absent from Bakta

Probiotic function	PGMDB gene / aliases	Bakta locus tags	Annotation
Acid and bile tolerance	ackA	FFBLOD_00357 FFBLOD_01239 FFBLOD_02256	Acetate kinase
	arlS	FFBLOD_03006	Histidine kinase
	fabH	FFBLOD_02924	Beta-ketoacyl-[acyl-carrier-protein] synthase III
	kup	FFBLOD_02863	putative potassium transport system protein Kup
	pepF	FFBLOD_01667 FFBLOD_02449	Oligoendopeptidase F oligoendopeptidase F
	pgm	FFBLOD_01911 FFBLOD_01912	Phosphoglucomutase phosphomannomutase, alpha beta alpha domain
	ppk	FFBLOD_00588	RNA degradosome polyphosphate kinase
	ppx3	FFBLOD_00587	exopolyphosphatase
	ptsI	FFBLOD_02305 FFBLOD_02427	General (non sugar-specific) component of the phosphoenolpyruvate-dependent sugar phosphotransferase system (sugar PTS). This major carbohydrate active-transport system catalyzes the phosphorylation of incoming sugar substrates concomitantly with their translocation across the cell membrane. Enzyme I transfers the phosphoryl group from phosphoenolpyruvate (PEP) to the phosphoryl carrier protein (HPr)
	rpsS	FFBLOD_02139	30S ribosomal protein S19
	usp5	FFBLOD_01173	universal stress protein
	yloU	FFBLOD_01783 FFBLOD_01784 FFBLOD_02679 FFBLOD_02680 FFBLOD_02947 FFBLOD_02970	Alkaline shock protein Asp23/Gls24 family envelope stress response protein Stress response regulator gls24-like protein
Acid stress	atpA	FFBLOD_00448	F0F1 ATP synthase subunit alpha
	atpB	FFBLOD_00452	F0F1 ATP synthase subunit A
	atpC	FFBLOD_00445	F0F1 ATP synthase subunit epsilon
	atpD	FFBLOD_00446	F0F1 ATP synthase subunit beta
	atpE	FFBLOD_00451	F0F1 ATP synthase subunit C
	atpF	FFBLOD_00450	F0F1 ATP synthase subunit B
	atpG	FFBLOD_00447	F0F1 ATP synthase subunit gamma
	atpH	FFBLOD_00449	ATP synthase F1 subunit delta
	clpB	FFBLOD_02724	ATP-dependent chaperone ClpB
	clpP/clpP	FFBLOD_00638	ATP-dependent Clp endopeptidase proteolytic subunit ClpP
	dltA	FFBLOD_02612	D-alanine--poly(phosphoribitol) ligase subunit DltA
	dltC	FFBLOD_02614	D-alanine--poly(phosphoribitol) ligase subunit DltC

Probiotic function	PGMDB gene / aliases	Bakta locus tags	Annotation
	dltD	FFBLOD_02615	D-alanyl-lipoteichoic acid biosynthesis protein DltD
	gadB	FFBLOD_01968	Belongs to the group II decarboxylase family
	gap	FFBLOD_00631	type I glyceraldehyde-3-phosphate dehydrogenase
	groL/groEL	FFBLOD_00674	chaperonin GroEL
	guaA	FFBLOD_02062	glutamine-hydrolyzing GMP synthase
	ldh	FFBLOD_00784	Belongs to the LDH MDH superfamily. LDH family
	nhaC	FFBLOD_01165	Na+/H+ antiporter NhaC
	pgi	FFBLOD_00557	glucose-6-phosphate isomerase
	plsC	FFBLOD_02583	1-acyl-sn-glycerol-3-phosphate acyltransferase
	pyk	FFBLOD_02727	pyruvate kinase
	recA	FFBLOD_00403	recombinase RecA
	relA	FFBLOD_02691	In eubacteria ppGpp (guanosine 3'-diphosphate 5-' diphosphate) is a mediator of the stringent response that coordinates a variety of cellular activities in response to changes in nutritional abundance
	tpiA	FFBLOD_00629	triose-phosphate isomerase
	uvrA	FFBLOD_00643 FFBLOD_01067 FFBLOD_02412	UvrABC system protein A excinuclease ABC subunit UvrA
	yjbM	FFBLOD_01004 FFBLOD_02452	GTP pyrophosphokinase GTP pyrophosphokinase family protein
	yphJ	FFBLOD_01847	decarboxylase
	yvgP	FFBLOD_00610	Sodium proton antiporter
	ywaC	FFBLOD_01004	RelA SpoT domain protein
Acid stress / Heat stress	clpC	FFBLOD_02125	Part of a stress-induced multi-chaperone system, it is involved in the recovery of the cell from heat-induced damage, in cooperation with DnaK, DnaJ and GrpE
Acid stress/ bile resistance	copA	FFBLOD_01082	P-type ATPase
	dnaK	FFBLOD_02609	molecular chaperone DnaK
	eno	FFBLOD_00628	phosphopyruvate hydratase
	pgk	FFBLOD_00630	Phosphoglycerate kinase
Acid stress/ bile resistance/temperature	dnaJ	FFBLOD_02610	molecular chaperone DnaJ
	groS/groES	FFBLOD_00675	co-chaperone GroES
	grpE	FFBLOD_02608	nucleotide exchange factor GrpE
	htrA	FFBLOD_01351	serine protease
	luxS	FFBLOD_01013	S-ribosylhomocysteine lyase
Adhesion	EpsC	FFBLOD_00129	serine O-acetyltransferase EpsC
	EpsG	FFBLOD_02192	EpsG family protein
	epsA	FFBLOD_01825	PAP2 superfamily

Probiotic function	PGMDB gene / aliases	Bakta locus tags	Annotation
	epsB	FFBLOD_02179 FFBLOD_02207	Chain length determinant protein biosynthesis protein
	exoA	FFBLOD_00612	exodeoxyribonuclease III
	fbpA/fbp	FFBLOD_02816	Fibronectin-binding protein
	lspA	FFBLOD_02821	signal peptidase II
	tuf	FFBLOD_02538	elongation factor Tu
Adhesion or host interaction	glnH	FFBLOD_02046	ABC transporter substrate-binding protein
	psaA	FFBLOD_00731 FFBLOD_00835	Belongs to the bacterial solute-binding protein 9 family
Adhesion/Temperature stress	hsp33/hslO	FFBLOD_00774	Hsp33 family molecular chaperone HslO
Adhesions	mapA	FFBLOD_00995	hydrolase, family 65, central catalytic
Antioxidant	mntH	FFBLOD_00921 FFBLOD_01022 FFBLOD_01139	Nramp family divalent metal transporter
	msrA	FFBLOD_02699 FFBLOD_02780	Peptide methionine sulfoxide reductase MsrA
	msrB	FFBLOD_02779	Peptide methionine sulfoxide reductase MsrB
	msrC	FFBLOD_00424	Histidine kinase
	ndh	FFBLOD_01036	NADH dehydrogenase, FAD-containing subunit
	npr	FFBLOD_00122 FFBLOD_02892	Pyridine nucleotide-disulphide oxidoreductase Pyridine nucleotide-disulphide oxidoreductase , dimerisation domain
	nrdH	FFBLOD_00696	Glutaredoxin-like protein NrdH
	tpx	FFBLOD_00420	thiol peroxidase
	trxA	FFBLOD_00382 FFBLOD_01019 FFBLOD_03040	Thiol-disulfide isomerase and thioredoxin thioredoxin
Bile resistance	argS	FFBLOD_02434	arginine--tRNA ligase
	cfa	FFBLOD_01740	Cyclopropane-fatty-acyl-phospholipid synthase
	clpE	FFBLOD_02302	Belongs to the ClpA ClpB family
	dps	FFBLOD_00599 FFBLOD_03186	DNA-binding ferritin-like protein (Oxidative damage protectant) DNA-binding ferritin-like protein, DPS family
	glf	FFBLOD_02200	UDP-galactopyranose mutase
	glnA	FFBLOD_02980	type I glutamate--ammonia ligase
	lpdA	FFBLOD_02518	dihydrolipoyl dehydrogenase
	nagB/gnp	FFBLOD_00603	Glucosamine-6-phosphate deaminase
	oppA	FFBLOD_01323 FFBLOD_01360 FFBLOD_01373 FFBLOD_01376 FFBLOD_01560 FFBLOD_01857 FFBLOD_01869 FFBLOD_02050 FFBLOD_02051 FFBLOD_02500	ABC-type oligopeptide transport system, periplasmic component
	pepO	FFBLOD_01929	Neutral endopeptidase

Probiotic function	PGMDB gene / aliases	Bakta locus tags	Annotation
	ponA/pbp1A	FFBLOD_02839	penicillin-binding protein 1A
	ppaC	FFBLOD_02777	inorganic pyrophosphatase
	pyrG	FFBLOD_00818	CTP synthase
	rplD	FFBLOD_02136	50S ribosomal protein L4
	rplE	FFBLOD_02147	50S ribosomal protein L5
	rplF	FFBLOD_02150	50S ribosomal protein L6
	rpsC	FFBLOD_02141	30S ribosomal protein S3
	rpsE	FFBLOD_02152	30S ribosomal protein S5
Bile resistance/Adhesion	srtA	FFBLOD_00809	Sortase (Surface protein transpeptidase)
Cell wall formation or adhesion	dacC	FFBLOD_01084 FFBLOD_01732 FFBLOD_01989 FFBLOD_01990	D-alanyl-D-alanine carboxypeptidase serine-type D-Ala-D-Ala carboxypeptidase
	dltX	FFBLOD_02611	D-Ala-teichoic acid biosynthesis protein
	murA	FFBLOD_00443 FFBLOD_00811	UDP-N-acetylglucosamine 1-carboxyvinyltransferase
	tagE3	FFBLOD_02461	Glycosyl transferases group 1
	ytgP	FFBLOD_01608 FFBLOD_03048	Polysaccharide biosynthesis protein
	ywqD	FFBLOD_02180	Capsular exopolysaccharide family
Cold stress	cspB	FFBLOD_02485	Cold shock protein
	cspC	FFBLOD_01874 FFBLOD_02485	CSD domain-containing protein Cold-shock DNA-binding protein family
	rnr	FFBLOD_00622	ribonuclease R
Exopolysaccharide production	cpsA/cpsB/cpsC/cpsD	FFBLOD_02179	Capsular polysaccharide biosynthesis protein CpsC
	epsF/epsG/epsH/epsI/epsJ	FFBLOD_02192	EpsG family protein
	epsK/epsL/tagH/tagG-like	FFBLOD_00597	Teichoic acids export ATP-binding protein TagH
	galE/galT/galK	FFBLOD_01915 FFBLOD_01916 FFBLOD_03137 FFBLOD_03138 FFBLOD_03139	UDP-glucose 4-epimerase GalE UDP-glucose--hexose-1-phosphate uridylyltransferase galactokinase
	galU	FFBLOD_00649	UTP--glucose-1-phosphate uridylyltransferase GalU
	glmS/glmM/glmU	FFBLOD_00605 FFBLOD_00606 FFBLOD_00829	bifunctional UDP-N-acetylglucosamine diphosphorylase/glucosamine-1-phosphate N-acetyltransferase GlmU glutamine--fructose-6-phosphate transaminase (isomerizing) phosphoglucosamine mutase
	manA/manB/manC	FFBLOD_02168	mannose-6-phosphate isomerase, class I
	pgmA/pgm	FFBLOD_01911 FFBLOD_01912	Phosphoglucomutase phosphomannomutase, alpha beta alpha domain
	wza/wzb/wzc	FFBLOD_00798	protein-tyrosine-phosphatase
GABA production	gabD	FFBLOD_01033 FFBLOD_01133	Belongs to the aldehyde dehydrogenase family
Heat-shock stress	hslR	FFBLOD_00780	RNA-binding S4 domain-containing protein

Probiotic function	PGMDB gene / aliases	Bakta locus tags	Annotation
Immunomodulation	dltB	FFBLOD_02613	D-alanyl-lipoteichoic acid biosynthesis protein DltB
	dltC	FFBLOD_02614	D-alanine--poly(phosphoribitol) ligase subunit DltC
	dltD	FFBLOD_02615	D-alanyl-lipoteichoic acid biosynthesis protein DltD
Intestinal persistence	celB	FFBLOD_01011 FFBLOD_01995	Permease IIC component
Ionic and heavy metal stress resistance	corA	FFBLOD_00157 FFBLOD_01044	CorA-like protein Mg2+ and Co2+ transporter
	czcD	FFBLOD_02106	Co/Zn/Cd efflux system component
	zntA	FFBLOD_00601 FFBLOD_00965 FFBLOD_01082 FFBLOD_01086	Cd(2+)-exporting ATPase P-type Cu(+) transporter
Nitrite stress	pgl	FFBLOD_01943	Carboxy-cis,cis-muconate cyclase
Osmotic stress	glpF	FFBLOD_01802 FFBLOD_01962 FFBLOD_02238	Glycerol uptake facilitator related permease Glycerol uptake facilitator related permease (Major Intrinsic Protein Family)
	opuCA	FFBLOD_01060	ABC transporter, ATP-binding protein
	opuCB	FFBLOD_01061	ABC transporter permease
Oxidative stress	gpo	FFBLOD_01792	Belongs to the glutathione peroxidase family
	katE	FFBLOD_02895	Catalase
	msrB	FFBLOD_02779	Peptide methionine sulfoxide reductase MsrB
	nox/nox2	FFBLOD_01020 FFBLOD_02298	NADH oxidase
	ywbN	FFBLOD_01057	Peroxidase
Polyamine production	aguA	FFBLOD_01510 FFBLOD_01512	agmatine deiminase
	aguD	FFBLOD_01509	Amino Acid
	potA/potB/potC/potD	FFBLOD_01954 FFBLOD_01955 FFBLOD_01956 FFBLOD_01957	Spermidine/putrescine ABC transporter permease protein Spermidine/putrescine import ATP-binding protein PotA Spermidine/putrescine-binding periplasmic protein
	potE	FFBLOD_00346 FFBLOD_00673 FFBLOD_01300 FFBLOD_01509 FFBLOD_01661 FFBLOD_01941 FFBLOD_02039 FFBLOD_02176 FFBLOD_02270	Agmatine/putrescine exchanger Amino acid transporter Glutamate/GABA antiporter
Proteases	clpL	FFBLOD_00143	ATP-dependent protease ClpL
	clpX	FFBLOD_02540	ATP-dependent Clp protease ATP-binding subunit ClpX
SCFA production	etfA/etfB	FFBLOD_01907	Electron transfer flavoprotein FAD-binding domain
	fumA/fumB/fumC	FFBLOD_00577	Fumarate hydratase class II
	gabT/gabD/4hbt	FFBLOD_01033 FFBLOD_01133	Belongs to the aldehyde dehydrogenase family

Probiotic function	PGMDB gene / aliases	Bakta locus tags	Annotation
	ldhD	FFBLOD_02586	Belongs to the D-isomer specific 2-hydroxyacid dehydrogenase family
	mdh	FFBLOD_00784 FFBLOD_02519	L-lactate dehydrogenase Lactate/malate dehydrogenase, NAD binding domain protein
	pduC/pduD/pduE	FFBLOD_02242 FFBLOD_02243 FFBLOD_02244	Propanediol dehydratase Propanediol dehydratase medium subunit PduD Propanediol dehydratase, small subunit
	pduL/pduW	FFBLOD_02249	Phosphate propanoyltransferase
	pduP	FFBLOD_02254	Aldehyde dehydrogenase family
	pduQ	FFBLOD_01804 FFBLOD_02255	Iron-containing alcohol dehydrogenase
	pta	FFBLOD_00617	phosphate acetyltransferase
	thl/atoB	FFBLOD_01507	Belongs to the thiolase family
Stress adaptation	lepA	FFBLOD_02616	translation elongation factor 4
Stress response	pbpX	FFBLOD_02473	Penicillin-binding Protein
Temperature stress	ctsR	FFBLOD_02124	Transcriptional regulator CtsR
	hrcA	FFBLOD_02607	heat-inducible transcriptional repressor HrcA
	hslV	FFBLOD_02770	HslVU peptidase proteolytic subunit
	htpX	FFBLOD_00796	zinc metalloprotease HtpX
Tryptophan metabolism	aroE	FFBLOD_01900	Shikimate dehydrogenase (NADP(+))
	fldA/fldB/fldC	FFBLOD_01108 FFBLOD_02286	Flavodoxin flavodoxin
	trpF	FFBLOD_01593	N-(5'-phosphoribosyl)anthranilate isomerase
Vitamin biosynthesis	bioY/bioM/bioN	FFBLOD_02912	Biotin transporter
	cbiA/cbiB/cbiC/cbiD/cbiE/cbiF/c	FFBLOD_00864 FFBLOD_00865 FFBLOD_00868 FFBLOD_00869 FFBLOD_00870 FFBLOD_00871 FFBLOD_00872 FFBLOD_00873 FFBLOD_00874 FFBLOD_00875 FFBLOD_00876 FFBLOD_00877	Catalyzes the ATP-dependent amidation of the two carboxylate groups at positions a and c of cobyrinate, using either L-glutamine or ammonia as the nitrogen source Cobalamin biosynthesis protein CbiG Cobalt chelatase (CbiK) Precorrin-8X methylmutase Protein-L-isoaspartate(D-aspartate) O-methyltransferase (PCMT) Tetrapyrrole (Corrin/Porphyrin) Methylases adenosylcobinamide-phosphate synthase CbiB cobalt-precorrin-5B (C(1))-methyltransferase CbiD
	coaA/coaB/coaC/coaD/coaE	FFBLOD_02060 FFBLOD_02526 FFBLOD_03030	dephospho-CoA kinase pantetheine-phosphate adenylyltransferase type I pantothenate kinase
	cobA/cobI/cobG/cobJ/cobM/cob	FFBLOD_00866 FFBLOD_00867	Tetrapyrrole (Corrin/Porphyrin) Methylases precorrin-6A reductase
	cobU/cobS/cobT	FFBLOD_00851 FFBLOD_00852	Joins adenosylcobinamide-GDP and alpha-ribazole to generate adenosylcobalamin (Ado-cobalamin). Also synthesizes adenosylcobalamin 5'-phosphate from adenosylcobinamide-GDP and alpha-ribazole 5'-phosphate bifunctional adenosylcobinamide kinase/adenosylcobinamide-phosphate guanylyltransferase
	folA	FFBLOD_02430 FFBLOD_02747	Dihydrofolate reductase

Probiotic function	PGMDB gene / aliases	Bakta locus tags	Annotation
	folC/folC1/folC2	FFBLOD_00418	tetrahydrofolate synthase
	menA/menG	FFBLOD_01424 FFBLOD_01894 FFBLOD_02226	1,4-Dihydroxy-2-naphtoate prenyltransferase 1,4-dihydroxy-2-naphthoate polyprenyltransferase Methyltransferase required for the conversion of demethylmenaquinol (DMKH2) to menaquinol (MKH2)
	menB	FFBLOD_01314	1,4-dihydroxy-2-naphthoyl-CoA synthase
	menE	FFBLOD_01313	o-succinylbenzoate--CoA ligase
	menH	FFBLOD_00746 FFBLOD_00954 FFBLOD_02037	Halo peroxidase 3 Hydrolase, alpha/beta domain protein Proline iminopeptidase
	nadD	FFBLOD_03018	nicotinate-nucleotide adenylyltransferase
	nadE	FFBLOD_00754	ammonia-dependent NAD(+) synthetase
	panE/ilvC-like	FFBLOD_01605	2-dehydropantoate 2-reductase
	pncA/pncB/nadV	FFBLOD_00755 FFBLOD_01114 FFBLOD_02910	Amidase nicotinate phosphoribosyltransferase
	ribA/ribBA	FFBLOD_02173	GTP cyclohydrolase-2
	ribD/ribG	FFBLOD_02171	bifunctional diaminohydroxyphosphoribosylaminopyrimidine deaminase/5-amino-6-(5-phosphoribosylamino)uracil reductase RibD
	ribE	FFBLOD_02172	Riboflavin synthase
	ribF/ribC	FFBLOD_02172 FFBLOD_02606	riboflavin biosynthesis protein RibF riboflavin synthase
	ribH	FFBLOD_02174	6,7-dimethyl-8-ribityllumazine synthase
	ribU	FFBLOD_02734	Mediates riboflavin uptake, may also transport FMN and roseoflavin. Probably a riboflavin-binding protein that interacts with the energy-coupling factor (ECF) ABC-transporter complex. Unlike classic ABC transporters this ECF transporter provides the energy necessary to transport a number of different substrates. The substrates themselves are bound by transmembrane, not extracytoplasmic soluble proteins
	thiD	FFBLOD_01585 FFBLOD_01691 FFBLOD_01881	Hydroxymethylpyrimidine/phosphomethylpyrimidine kinase pyridoxal kinase
	thiE	FFBLOD_01692	Thiamine-phosphate synthase
	thiM/thiN/thiT	FFBLOD_01690 FFBLOD_02949	hydroxyethylthiazole kinase thiamine diphosphokinase
	thiS/thiF/thiI	FFBLOD_00421	putative tRNA sulfurtransferase

Gut-Brain Axis Markers: SRR30409709

Detected genes grouped by Metabolite/Trait; repeated logic and interpretation shown once

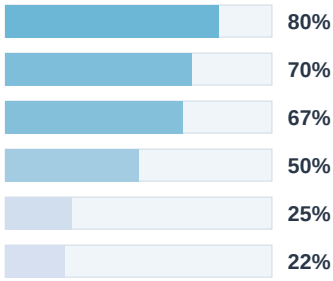
Metabolite/Trait	Markers	Bakta loci	KO/EC hints	Marker logic	Interpretation
GABA	gadB gadA	FFBLOD_01968 FFBLOD_01299	EC:4.1.1.15 KO:K01580 KO:ATP synthase complex	Full module if gadB or gadA plus gadC; partial if only gadB/gadA Presence of glutamate decarboxylase genes ATP synthase plus acid-stress systems	Potential glutamate-to-GABA conversion and GABA export/import GABA-related acid resistance and possible GABA formation Potential acid survival, indirectly supporting gut activity
GABA; acid tolerance	gadB gadA atpA atpB atpC atpD atpE atpF atpG	FFBLOD_01968 FFBLOD_01299 FFBLOD_00448 FFBLOD_00452 FFBLOD_00445 FFBLOD_00446 FFBLOD_00451 FFBLOD_00450 FFBLOD_00447	EC:4.1.1.15 KO:K01580 KO:ATP synthase complex	Full module if gadB or gadA plus gadC; partial if only gadB/gadA Presence of glutamate decarboxylase genes ATP synthase plus acid-stress systems	Potential glutamate-to-GABA conversion and GABA export/import GABA-related acid resistance and possible GABA formation Potential acid survival, indirectly supporting gut activity
Gastric acid survival	gadB atpA atpB atpC atpD atpE atpF atpG	FFBLOD_01968 FFBLOD_00448 FFBLOD_00452 FFBLOD_00445 FFBLOD_00446 FFBLOD_00451 FFBLOD_00450 FFBLOD_00447	EC:4.1.1.15 KO:K01580 KO:ATP synthase complex	Full module if gadB or gadA plus gadC; partial if only gadB/gadA Presence of glutamate decarboxylase genes ATP synthase plus acid-stress systems	Potential glutamate-to-GABA conversion and GABA export/import GABA-related acid resistance and possible GABA formation Potential acid survival, indirectly supporting gut activity
GABA shunt	gabD	FFBLOD_01033 FFBLOD_01133	EC:2.6.1.19 EC:1.2.1.16 KO:K07250 KO:K00135	gabT plus gabD indicates GABA catabolism	Potential GABA utilization via succinate pathway
Indole-3-lactate and related derivatives	hydroxyacid_dehydrog	FFBLOD_01226 FFBLOD_02019 FFBLOD_02341 FFBLOD_02586	EC:2.6.1.57-like EC:1.1.1.x	Aromatic aminotransferase plus hydroxyacid/lactate dehydrogenase-like enzyme Aromatic aminotransferase plus hydroxyacid dehydrogenase-like enzyme	Putative aromatic lactate or indole-lactate related metabolism Putative aromatic lactate production potential
Phenyllactate; hydroxyphenyllactate	hydroxyacid_dehydrog	FFBLOD_01226 FFBLOD_02019 FFBLOD_02341 FFBLOD_02586	EC:2.6.1.57-like EC:1.1.1.x	Aromatic aminotransferase plus hydroxyacid/lactate dehydrogenase-like enzyme Aromatic aminotransferase plus hydroxyacid dehydrogenase-like enzyme	Putative aromatic lactate or indole-lactate related metabolism Putative aromatic lactate production potential
Chorismate precursor	aroE	FFBLOD_01900	KO:K00800 K01735 K01736 K03785 K03786	Full shikimate core if aroA-E present	Potential chorismate and aromatic amino acid precursor biosynthesis
Putrescine	aguA	FFBLOD_01510 FFBLOD_01512	EC:3.5.3.11 EC:4.1.1.19 EC:4.1.1.17	Agmatine route genes downstream of speA Presence of arginine/agmatine or ornithine route	Potential putrescine formation via agmatine Potential putrescine production
Polyamine uptake/export	potA potB potC potD	FFBLOD_01957 FFBLOD_01956 FFBLOD_01955 FFBLOD_01954	KO:K02052-like K02053-like K02054-like K02055-like	Transporters support polyamine exchange but not synthesis	Potential polyamine transport

Metabolite/Trait	Markers	Bakta loci	KO/EC hints	Marker logic	Interpretation
Cadaverine	cadA	FFBLOD_00601 FFBLOD_00965	EC:4.1.1.18 KO:K01582-like	cadA/ldcC indicates lysine decarboxylation Presence of lysine decarboxylase cluster	Potential cadaverine production
Vitamin B9 / folate	folC folA	FFBLOD_00418 FFBLOD_02430 FFBLOD_02747	KO:K01495 K01633 K00940 K00796 K01934 K00287	Full if core folate genes present; partial otherwise	Potential folate biosynthesis
Vitamin B2 / riboflavin	ribA ribD ribE ribH	FFBLOD_02173 FFBLOD_02171 FFBLOD_02172 FFBLOD_02174	KO:K02858 K14652 K00082 K00794 K00793	Full riboflavin module if ribA/B/D/E/H present	Potential riboflavin biosynthesis
Vitamin B1 / thiamine	thiD thiE	FFBLOD_01585 FFBLOD_01691 FFBLOD_01881 FFBLOD_01692	KO:K03147 K00941 K00788-like	Full pathway if thiazole and pyrimidine branches plus thiE present	Potential thiamine biosynthesis
Vitamin B12 / cobalamin	cobA cobB cobC cobD cobK cobQ cobS cobU	FFBLOD_00866 FFBLOD_00891 FFBLOD_00850 FFBLOD_00878 FFBLOD_00867 FFBLOD_00460 FFBLOD_00859 FFBLOD_00851 FFBLOD_00852	KO:cob/cbi modules	High only if near-complete cob/cbi pathway; transport alone is not synthesis	Potential cobalamin biosynthesis or uptake
Lactate	ldh ldhD	FFBLOD_00784 FFBLOD_02586	EC:1.1.1.27 KO:K00016	Presence of lactate dehydrogenase genes	Potential lactate fermentation
Acetate	pta ackA	FFBLOD_00617 FFBLOD_00357 FFBLOD_01239 FFBLOD_02256	EC:2.3.1.8 EC:2.7.2.1 KO:K00625 K00925 KO:K00656 K04072	pta plus ackA indicates acetate route Presence of pyruvate formate lyase and alcohol dehydrogenase	Potential acetate production Potential mixed-acid fermentation products
Formate; ethanol	pta ackA adhE	FFBLOD_00617 FFBLOD_00357 FFBLOD_01239 FFBLOD_02256 FFBLOD_01033 FFBLOD_01133 FFBLOD_01249 FFBLOD_01275 FFBLOD_02254	EC:2.3.1.8 EC:2.7.2.1 KO:K00625 K00925 KO:K00656 K04072	pta plus ackA indicates acetate route Presence of pyruvate formate lyase and alcohol dehydrogenase	Potential acetate production Potential mixed-acid fermentation products
Propionate	pduP pduL	FFBLOD_02254 FFBLOD_02249	KO:pathway-specific	Presence of a recognized propionate pathway	Potential propionate production
Exopolysaccharide production	epsA epsB epsC	FFBLOD_01825 FFBLOD_02179 FFBLOD_02207 FFBLOD_00129	NA	Presence of eps cluster core genes	Potential EPS biosynthesis and immunomodulatory support
Mucosal persistence	srtA	FFBLOD_00809	PFAM:LPXTG_sortas	EPS plus adhesion/surface protein markers	Potential mucosal persistence and host interaction
Reactive oxygen stress resistance	katA katE nox trxA trxB	FFBLOD_02895 FFBLOD_01020 FFBLOD_02298 FFBLOD_00382 FFBLOD_01019 FFBLOD_03040 FFBLOD_00648 FFBLOD_02003	EC:1.15.1.1 EC:1.11.1.6	Presence of multiple oxidative stress genes	Potential survival in oxidative gut/food environment
Bile survival	choloylglycine_hydrola	FFBLOD_00570	EC:3.5.1.24	Presence of bile salt hydrolase and/or bile resistance genes	Potential bile tolerance and gut persistence

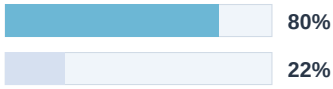
KEGGDecoder Functional Completeness

Sample SRR30409709 - all 15 detected functions; completeness from 0 to 100%

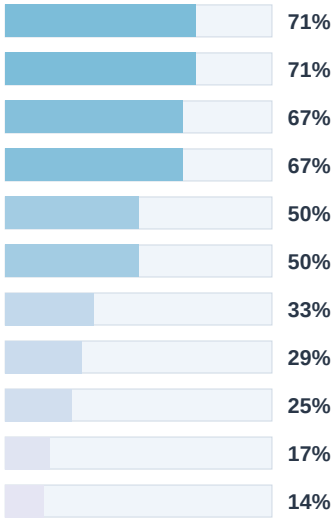
Probiotic-relevant detected functions



Vitamins & cofactors



Other detected functions



Carbohydrates & fermentation

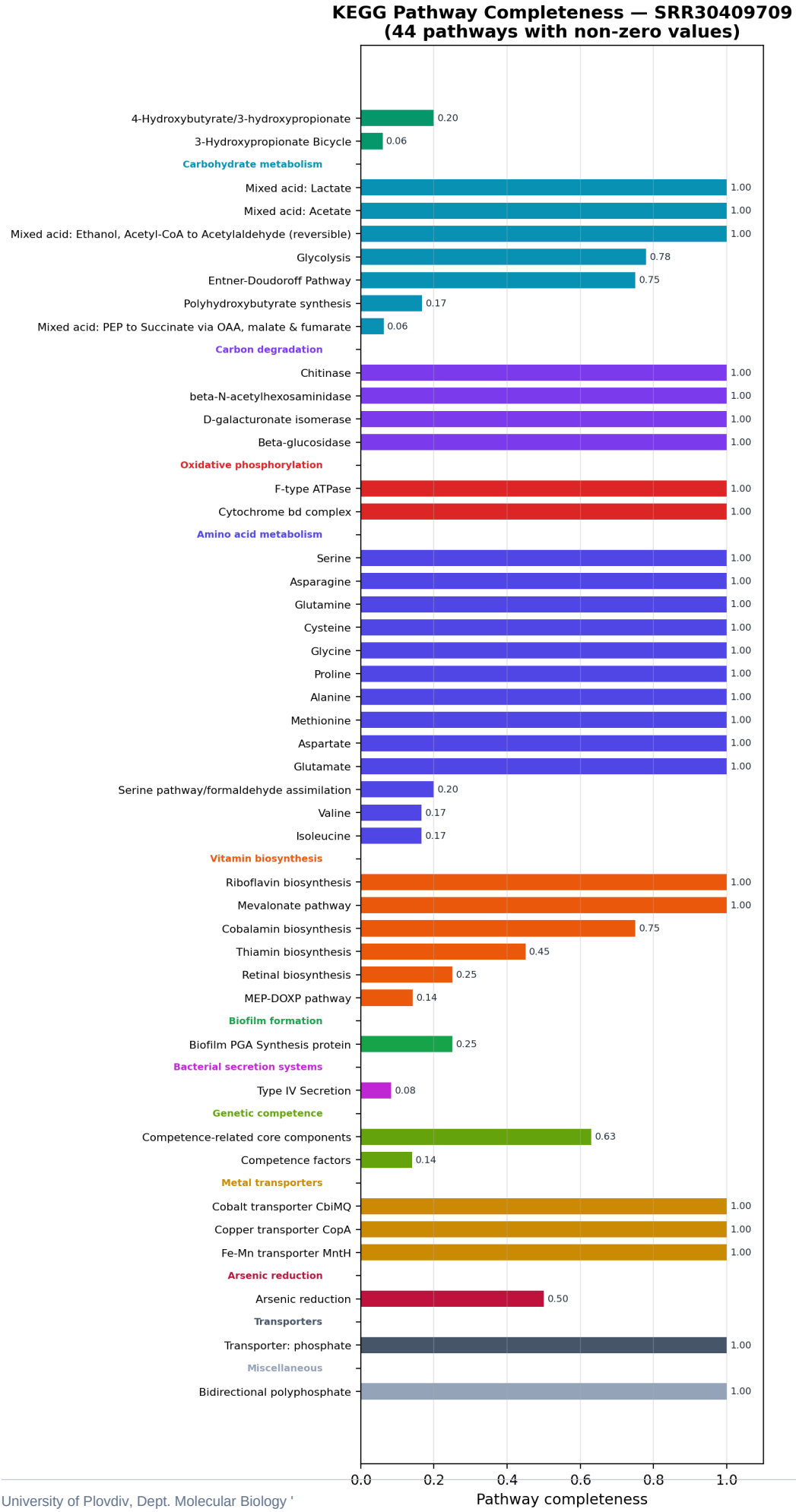


Transport, stress & interaction



KEGGaNOG Pathway Completeness (from eggNOG)

Sample SRR30409709 ? non-zero KEGG pathways from eggNOG-mapper annotations



Nutrient Uptake Potential (KEGG TC) - SRR30409709

Transporter classification from eggNOG-mapper; probiotic relevance annotation

Category	TC class	Transporter family	Probiotic relevance	Unique TCs
Carbohydrate uptake	4.A.1	PTS systems	Glucose, fructose, mannose, cellobiose uptake; prebiotic utilization	0
	3.A.1	ABC sugar transporters	Oligosaccharide and sugar uptake	51
	2.A.1	MFS transporters	Lactose, galactose and carbohydrate transport	19
Oligosaccharide uptake	3.A.1.5	Oligosaccharide ABC transporters	FOS, GOS, XOS, raffinose-family oligosaccharide utilization	5
Peptide uptake	3.A.1.17	Opp/Dpp peptide transporters	Peptide nutrition, gut adaptation and colonization	1
Amino acid & polyamine uptake	2.A.3	APC family transporters	Amino acid uptake, polyamine metabolism	6
Stress resistance	2.A.12	Glycine betaine/choline transporters	Osmotic stress tolerance	0
	2.A.66	MATE transporters	Bile tolerance and multidrug efflux	2
	2.A.6	RND efflux systems	Toxin and stress resistance	0
Ion homeostasis	3.A.3	ATP-driven ion transporters	pH regulation and ion balance	4

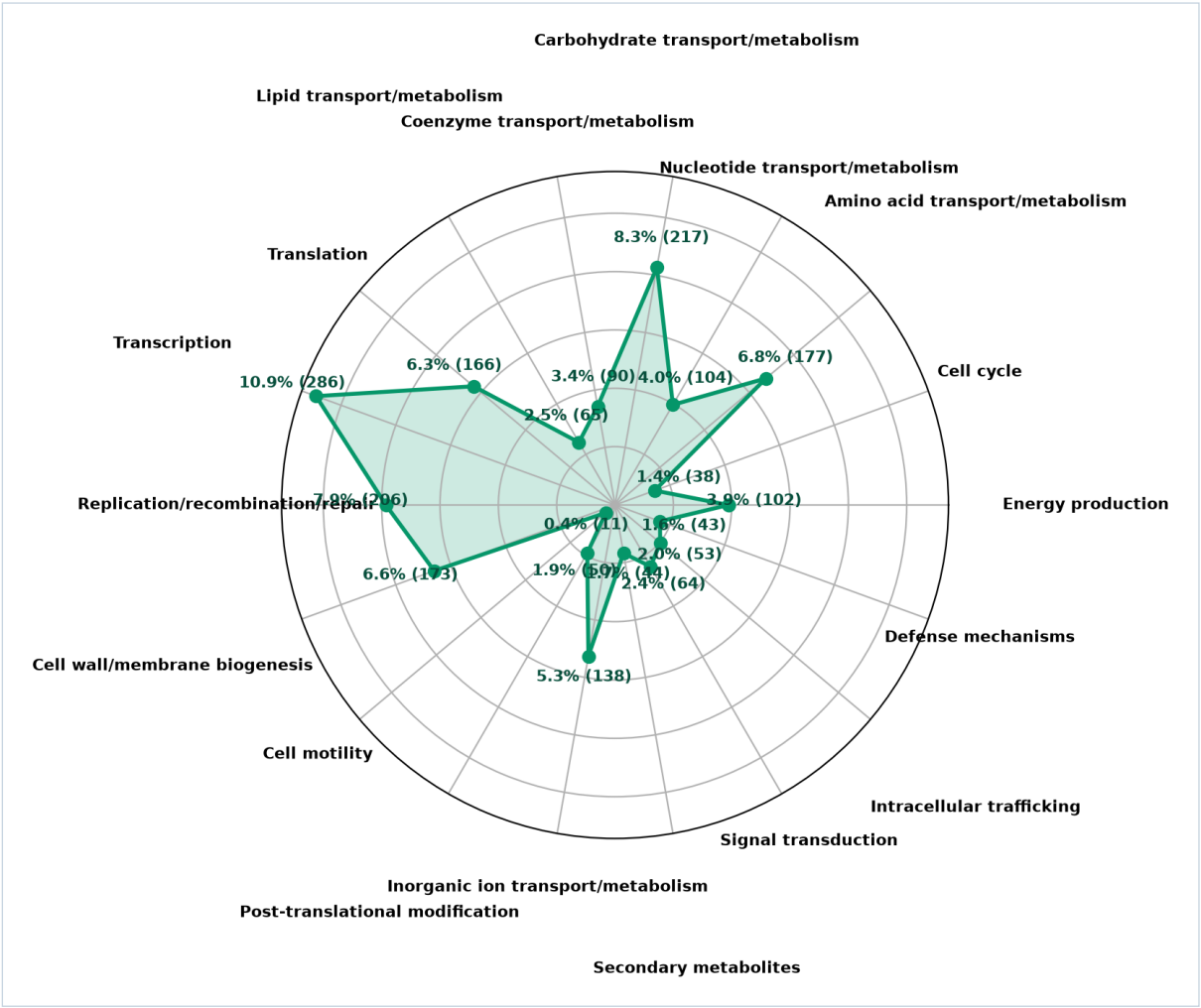
EC Enzyme Classification - SRR30409709

Enzyme Commission classes from eggNOG-mapper annotations

EC class	Subcategory	EC prefix	Description	Unique ECs
EC 1	Dehydrogenases	1.1.*	Acting on CH-OH groups	32
	Aldehyde dehydrogenases	1.2.*	Acting on aldehydes	10
	Oxidases	1.3.*, 1.4.*	Oxidation reactions	12
	Peroxidases/Catalases	1.11.*	Oxidative stress defense	4
	Superoxide dismutases	1.15.1.*	Reactive oxygen detoxification	0
EC 2	Kinases	2.7.*	Phosphoryl transfer	78
	Glycosyltransferases	2.4.*	EPS and cell wall synthesis	21
	Acyltransferases	2.3.*	Lipid and SCFA metabolism	23
	Aminotransferases	2.6.*	Amino acid metabolism	3
	Methyltransferases	2.1.*	Methylation reactions	43
EC 3	Glycosidases	3.2.*	Carbohydrate degradation	23
	Peptidases	3.4.*	Protein degradation	22
	Amidases	3.5.*	Nitrogen metabolism	25
	Phosphatases	3.1.3.*, 3.6.*	Phosphate metabolism	40
	Esterases/Lipases	3.1.1.*	Lipid degradation	3
EC 4	Decarboxylases	4.1.*	GABA and polyamine pathways	23
	Aldolases	4.1.2.*	Central carbon metabolism	7
	Ammonia-lyases	4.3.*	Amino acid metabolism	2
EC 5	Isomerases	5.3.*	Sugar interconversion	12
	Epimerases/Racemases	5.1.*	Stereochemical conversions	14
	Mutases	5.4.*	Metabolic rearrangements	17
EC 6	C-O bond ligases	6.1.*	Aminoacyl-tRNA synthetases	20
	C-N bond ligases	6.3.*	Peptide/cofactor biosynthesis	33
	DNA ligases	6.5.*	Genome maintenance	1
EC 7	Hydrogen ion translocases	7.1.*	Proton motive force	0
	Inorganic cation translocases	7.2.*	Ion homeostasis	0
	ABC transport translocases	7.6.*	ATP-dependent transport	0

COG Functional Categories

Sample SRR30409709 - COG category distribution from eggNOG-mapper



GenomeSPOT Growth Condition Predictions: SRR30409709

Predicted oxygen tolerance, temperature, salinity and pH from genome features

Condition	Predicted value	Error	Units
Oxygen	tolerant	0.999361896610799	probability
Ph Max	6.58	2.12	pH
Ph Min	4.37	0.85	pH
Ph Optimum	5.37	1.42	pH
Salinity Max	5.76	3.42	% w/v NaCl
Salinity Min	2.38	2.29	% w/v NaCl
Salinity Optimum	2.70	2.21	% w/v NaCl
Temperature Max	33.32	5.82	C
Temperature Min	8.79	6.21	C
Temperature Optimum	20.14	7.27	C

gutSMASH Metabolic Clusters: SRR30409709

Identified primary metabolite regions from gutSMASH index.html

Region	Type	Class	From	To	Most similar known cluster	Similarity
Region 5.2	Arginine2_Hcarbonate	Other	1,506,229	1,532,116	Arginine to hydrogen carbonate P. aeruginosa (ARG)	100%
Region 5.3	EUT_pathway, gallic_acid_met	Aromatic	1,537,827	1,588,997	Ethanolamine degradation S. typhimurium (EUT)	63%
Region 5.4	pdu	SCFA-other	1,964,902	2,001,589	Propanediol degradation S. enterica (PDU)	70%

GO Enrichment: SRR30409709

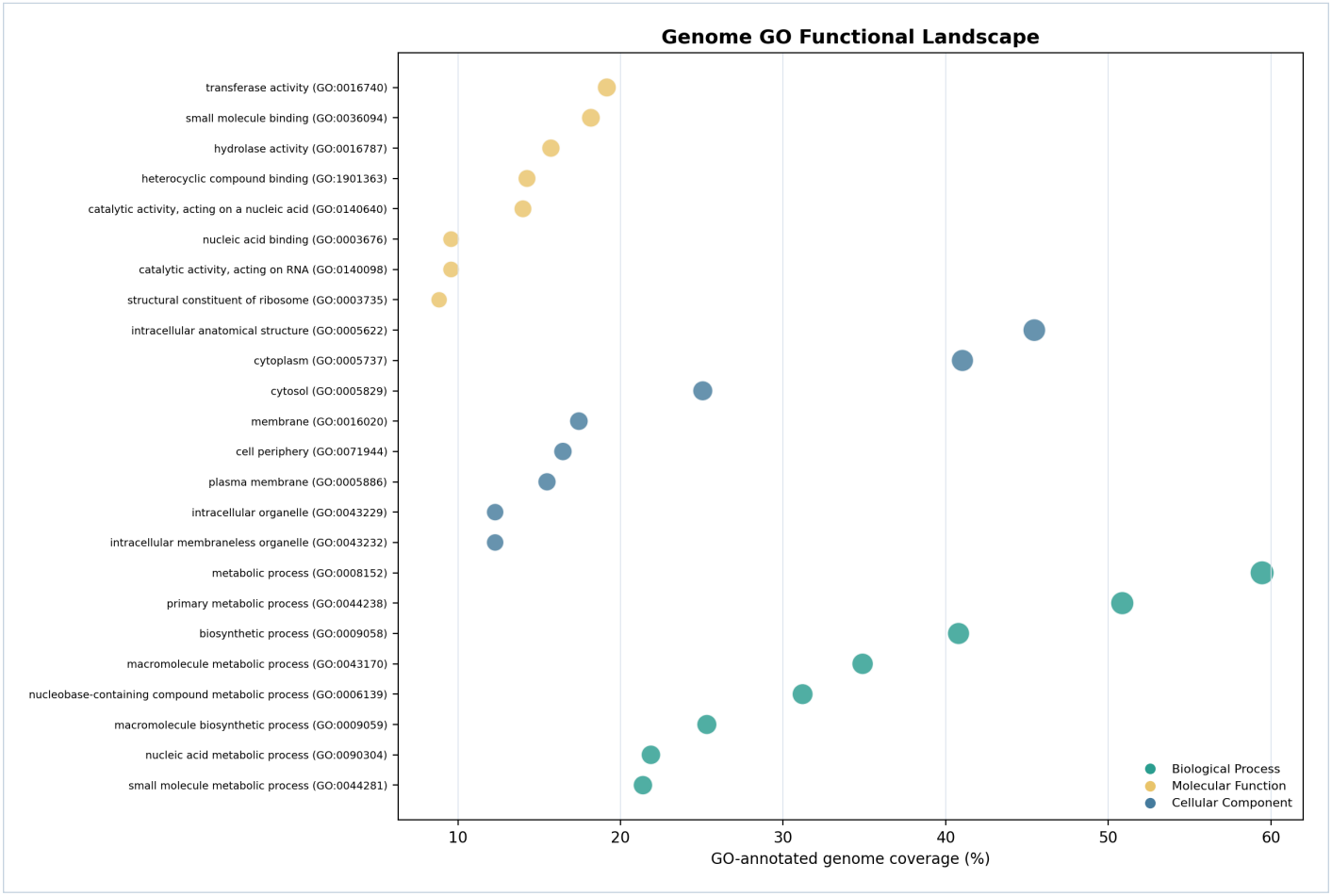
Significantly enriched GO terms (FDR < 0.05)

GO ID	Name	Namespace	Fold Enrichment	FDR
GO:0006805	xenobiotic metabolic process	biological_process	3.957	0.0001
GO:0009150	purine ribonucleotide metabolic process	biological_process	3.561	0.0005
GO:0006163	purine nucleotide metabolic process	biological_process	3.131	0.0006
GO:0072521	purine-containing compound metabolic process	biological_process	3.015	0.0007
GO:0006164	purine nucleotide biosynthetic process	biological_process	3.481	0.0007
GO:0072522	purine-containing compound biosynthetic process	biological_process	3.307	0.0015
GO:0046034	ATP metabolic process	biological_process	4.522	0.0022
GO:0006754	ATP biosynthetic process	biological_process	4.522	0.0022
GO:0009259	ribonucleotide metabolic process	biological_process	2.968	0.0024
GO:0009152	purine ribonucleotide biosynthetic process	biological_process	3.498	0.0024
GO:0019693	ribose phosphate metabolic process	biological_process	2.849	0.0042
GO:0009168	purine ribonucleoside monophosphate biosynthetic process	biological_process	3.816	0.0043
GO:0009127	purine nucleoside monophosphate biosynthetic process	biological_process	3.816	0.0043
GO:0009145	purine nucleoside triphosphate biosynthetic process	biological_process	4.07	0.0046
GO:0009206	purine ribonucleoside triphosphate biosynthetic process	biological_process	4.07	0.0046
GO:0009201	ribonucleoside triphosphate biosynthetic process	biological_process	4.07	0.0046
GO:0009205	purine ribonucleoside triphosphate metabolic process	biological_process	4.07	0.0046
GO:0005618	cell wall	cellular_component	4.452	0.0048
GO:0030312	external encapsulating structure	cellular_component	4.452	0.0048
GO:1901135	carbohydrate derivative metabolic process	biological_process	2.212	0.0068
GO:0009126	purine nucleoside monophosphate metabolic process	biological_process	3.522	0.0069
GO:0009167	purine ribonucleoside monophosphate metabolic process	biological_process	3.522	0.0069
GO:0006753	nucleoside phosphate metabolic process	biological_process	2.337	0.0072
GO:0009142	nucleoside triphosphate biosynthetic process	biological_process	3.7	0.0087
GO:0006091	generation of precursor metabolites and energy	biological_process	3.7	0.0087
GO:0009144	purine nucleoside triphosphate metabolic process	biological_process	3.7	0.0087
GO:0019363	pyridine nucleotide biosynthetic process	biological_process	3.957	0.0106
GO:0019359	nicotinamide nucleotide biosynthetic process	biological_process	3.957	0.0106
GO:0044281	small molecule metabolic process	biological_process	1.754	0.0106
GO:0009117	nucleotide metabolic process	biological_process	2.326	0.0106
GO:0072525	pyridine-containing compound biosynthetic process	biological_process	3.957	0.0106
GO:0009156	ribonucleoside monophosphate biosynthetic process	biological_process	3.271	0.0108
GO:0033865	nucleoside bisphosphate metabolic process	biological_process	5.088	0.0108

GO ID	Name	Namespace	Fold Enrichment	FDR
GO:0034032	purine nucleoside bisphosphate metabolic process	biological_process	5.088	0.0108
GO:0033875	ribonucleoside bisphosphate metabolic process	biological_process	5.088	0.0108
GO:1901137	carbohydrate derivative biosynthetic process	biological_process	2.261	0.0122
GO:0046390	ribose phosphate biosynthetic process	biological_process	2.798	0.0122
GO:0009260	ribonucleotide biosynthetic process	biological_process	2.798	0.0122
GO:0055086	nucleobase-containing small molecule metabolic process	biological_process	2.162	0.0179
GO:0009124	nucleoside monophosphate biosynthetic process	biological_process	3.053	0.0186
GO:0072524	pyridine-containing compound metabolic process	biological_process	3.561	0.0220
GO:0046496	nicotinamide nucleotide metabolic process	biological_process	3.561	0.0220
GO:0009141	nucleoside triphosphate metabolic process	biological_process	3.131	0.0296
GO:0046032	ADP catabolic process	biological_process	4.24	0.0321
GO:0009134	nucleoside diphosphate catabolic process	biological_process	4.24	0.0321
GO:0009137	purine nucleoside diphosphate catabolic process	biological_process	4.24	0.0321
GO:0046031	ADP metabolic process	biological_process	4.24	0.0321
GO:0072526	pyridine-containing compound catabolic process	biological_process	4.24	0.0321
GO:0009261	ribonucleotide catabolic process	biological_process	4.24	0.0321
GO:0009154	purine ribonucleotide catabolic process	biological_process	4.24	0.0321
GO:0006090	pyruvate metabolic process	biological_process	4.24	0.0321
GO:0042364	water-soluble vitamin biosynthetic process	biological_process	4.24	0.0321
GO:0042866	pyruvate biosynthetic process	biological_process	4.24	0.0321
GO:0008233	peptidase activity	molecular_function	4.24	0.0321
GO:0009110	vitamin biosynthetic process	biological_process	4.24	0.0321
GO:0009191	ribonucleoside diphosphate catabolic process	biological_process	4.24	0.0321
GO:0006096	glycolytic process	biological_process	4.24	0.0321
GO:0019364	pyridine nucleotide catabolic process	biological_process	4.24	0.0321
GO:1901293	nucleoside phosphate biosynthetic process	biological_process	2.281	0.0321
GO:0009165	nucleotide biosynthetic process	biological_process	2.281	0.0321
GO:0009181	purine ribonucleoside diphosphate catabolic process	biological_process	4.24	0.0321
GO:0140096	catalytic activity, acting on a protein	molecular_function	3.237	0.0354

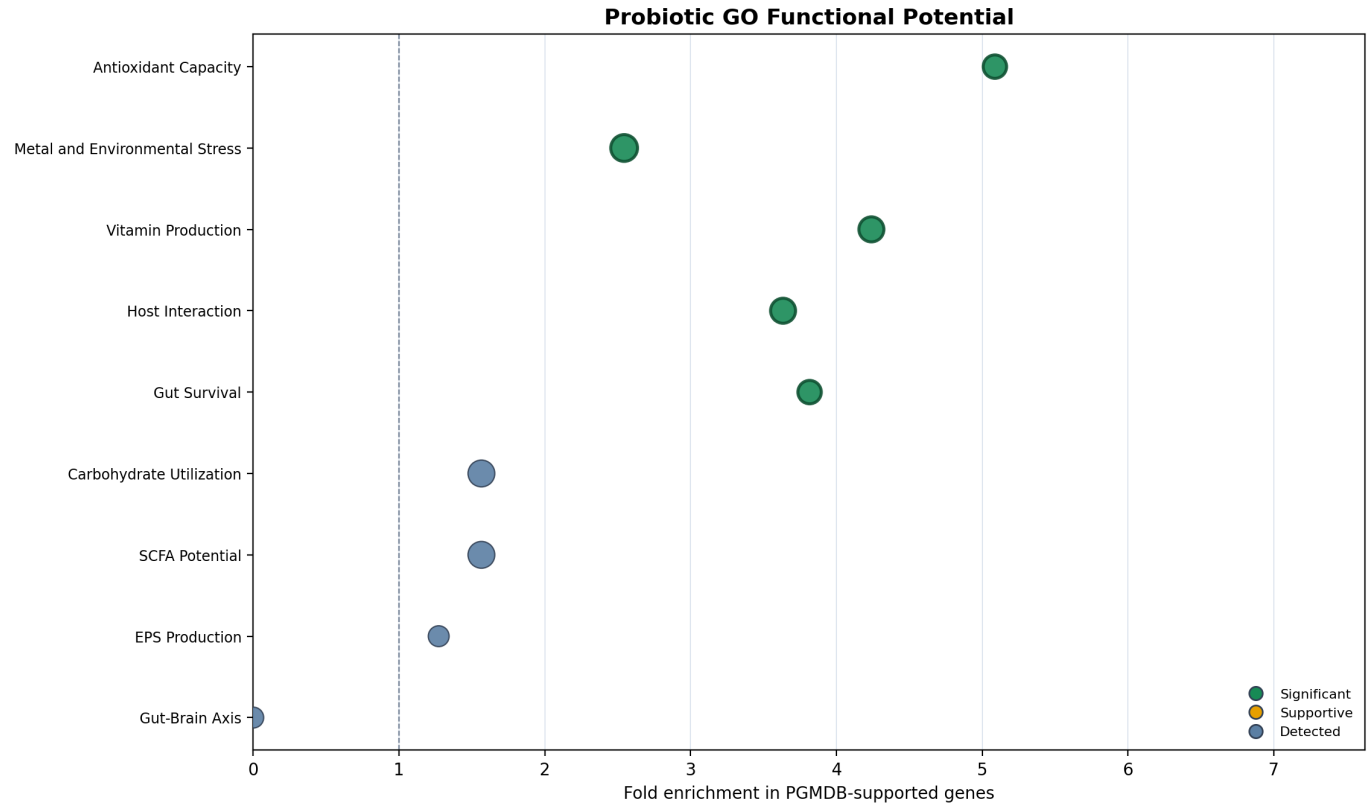
Genome GO Functional Landscape: SRR30409709

Bubble size represents gene count; position represents genome coverage; color represents GO branch



Probiotic GO Categories: SRR30409709

Category-level analysis: position is fold enrichment, bubble size is unique PGMDB-supported genes, and color is evidence tier



Results Directory Structure

Sample SRR30409709 - main output files from each analysis tool

data/results/SRR30409709/		
--	**FastQC/**	
|	+-- no expected output found	
OK	**Porechop/**	
|	+-- qc/porechop/SRR30409709.json	ONT adapter-trimming summary
OK	**Rasusa/**	
|	+-- qc/rasusa/SRR30409709_ONT.tsv	Read subsampling status
OK	**Flye/**	
|	+-- flye/assembly.fasta	ONT draft assembly
OK	**Medaka/**	
|	+-- medaka/consensus.fasta	Polished ONT assembly
OK	**QUAST/**	
|	+-- quast/report.tsv	Assembly quality statistics
OK	**Mash/**	
|	+-- mash/top_hits.tsv	RefSeq species identification
OK	**sourmash/**	
|	+-- sourmash/gather.with-lineages.csv	GTDB taxonomy classification
OK	**FastANI/**	
|	+-- fastani/SRR30409709_fastani.tsv	ANI comparisons
|	+-- fastani/SRR30409709_ani_heatmap.png	ANI heatmap
OK	**BUSCO/**	
|	+-- busco/busco_summary.txt	Genome completeness assessment
OK	**Bakta full/**	
|	+-- bakta/SRR30409709.gff3	Genome annotation
|	+-- bakta/SRR30409709.faa	Protein sequences
|	+-- bakta/SRR30409709.tsv	Feature table
|	+-- bakta/SRR30409709.png	Genome circular plot
OK	**AMRFinderPlus/**	
|	+-- amrfinder/SRR30409709.tsv	AMR gene scan
OK	**ABRicate CARD/MEGARes/**	
|	+-- abricate/SRR30409709_card.tsv	CARD AMR scan
|	+-- abricate/SRR30409709_megares.tsv	MEGARes AMR scan
OK	**PlasmidFinder/**	
|	+-- plasmidfinder/SRR30409709.tsv	Plasmid replicon detection
OK	**VFDB/**	
|	+-- vfdb/SRR30409709.tsv	Virulence-factor scan
OK	**geNomad/**	
|	+-- genomad/assembly_summary/assembly_plasmid_summary.tsv	Plasmid classification
|	+-- genomad/assembly_summary/assembly_virus_summary.tsv	Virus/provirus classification
OK	**MOB-recon/**	
|	+-- mobsuite/contig_report.txt	Per-contig plasmid classification
|	+-- mobsuite/mobtyper_results.txt	Reconstructed plasmid summary
OK	**eggNOG-mapper/**	
|	+-- eggnog/SRR30409709.emapper.annotations	Functional annotations
|	+-- eggnog/SRR30409709_transport.tsv	Transporter classification
|	+-- eggnog/SRR30409709_ec.tsv	Enzyme classes
|	+-- eggnog/SRR30409709_cog_radar.png	COG radar chart
OK	**KofamScan/**	
|	+-- kofamscan/SRR30409709.tsv	KEGG Orthology annotations
OK	**KEGGDecoder/**	
|	+-- keggdecoder/pathway_completeness.tsv	Pathway completeness
|	+-- keggdecoder/pathway_completeness.svg	Pathway heatmap
OK	**dbCAN v5/**	
|	+-- dbcan/overview.tsv	CAZyme annotations
|	+-- dbcan/cgc/cgc_with_substrates.tsv	CGC substrate predictions
OK	**PGMDB/**	
|	+-- pgmdb/summary.tsv	Probiotic marker summary
|	+-- pgmdb/hits.tsv	Probiotic marker details
|	+-- pgmdb/pgmdb_functional_potential.png	PGMDB gene coverage
|	+-- pgmdb/pgmdb_functional_potential.tsv	PGMDB gene coverage data
OK	**GO Probiotic/**	
|	+-- goprobioticprofiler/general_go_bubble.png	Genome GO functional landscape
|	+-- goprobioticprofiler/probiotic_go_bubble.png	Probiotic GO categories
|	+-- goprobioticprofiler/probiotic_go_categories.tsv	Probiotic GO category evidence
|	+-- goprobioticprofiler/probiotic_go_terms.tsv	Deduplicated probiotic GO term evidence
OK	**Gut-Brain markers/**	
|	+-- gut_brain_markers.tsv	Gut-brain axis marker table
OK	**KEGGaNOG/**	
|	+-- kegganog/SRR30409709_pathways.tsv	eggNOG pathway completeness
OK	**GenomeSPOT/**	
|	+-- genomespot/SRR30409709.predictions.tsv	Growth condition predictions

Results Directory Structure - continued

OK	**gutSMASH/**	
| +-- gutsmash/index.html	Gut metabolic cluster report	
OK	**BAGEL4/**	
| +-- bagel4/bacteriocin_clusters.tsv	Bacteriocin and RiPP clusters	
OK	**antiSMASH/**	
| +-- antismash/index.html	Biosynthetic cluster report	
| +-- antismash/SRR30409709_regions.tsv	Region summary	
OK	**pyCirclize genome map/**	
| +-- genome_map/SRR30409709_genome_map.png	Integrated bacterial genome map	
| +-- genome_map/SRR30409709_genome_map.svg	Editable vector genome map	